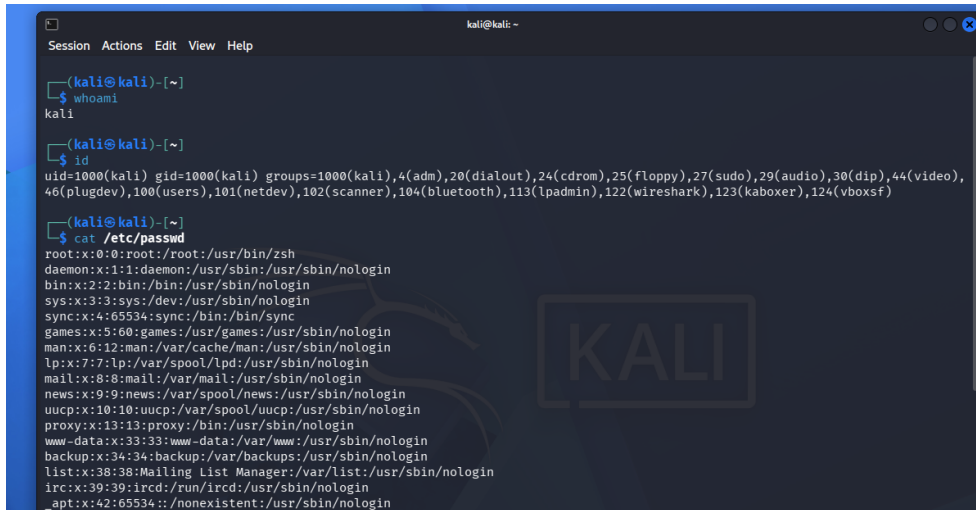


Linux Task 02: Users, Groups & File Permissions

Part A: Understanding Users

Run the following commands:

- whoami
- id
- cat /etc/passwd



```
kali@kali ~  
$ whoami  
kali  
$ id  
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),102(scanner),104(bluetooth),113(lpadmin),122(wireshark),123(kaboxer),124(vboxsf)  
$ cat /etc/passwd  
root:x:0:0:root:/root:/usr/bin/zsh  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin  
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
```

Answer the following:

- **What is your current username?**
Kali
- **What is UID?**
UID stands for User ID, it is a unique number given to a user
- **What is GID?**
GID stands for Group ID. It identifies the primary group to which a user belongs
- **What information does /etc/passwd contain?**
The file /etc/passwd contains information about all user accounts on the system, including username, UID, GID, User description Home idirectory etc.

Part B: Create Users & Groups

```
(kali㉿kali)-[~]
$ sudo groupadd interns
[sudo] password for kali:

(kali㉿kali)-[~]
$ sudo groupadd cyberteam

(kali㉿kali)-[~]
$ sudo useradd -m student1

(kali㉿kali)-[~]
$ sudo useradd -m student2

(kali㉿kali)-[~]
$ sudo useradd -m student3

(kali㉿kali)-[~]
$ sudo passwd student1
New password:
Retype new password:
passwd: password updated successfully

(kali㉿kali)-[~]
$ sudo passwd student2
New password:
Retype new password:
passwd: password updated successfully

(kali㉿kali)-[~]
$ sudo passwd student3
New password:
Retype new password:
passwd: password updated successfully

(kali㉿kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ sudo usermod -aG interns student1

(kali㉿kali)-[~]
$ sudo usermod -aG interns student2

(kali㉿kali)-[~]
$ sudo usermod -aG cyberteam student3

(kali㉿kali)-[~]
$ groups
kali adm dialout cdrom floppy sudo audio dip video plugdev users netdev scanner bluetooth lpadmin wireshark kaboxer vboxs

(kali㉿kali)-[~]
$ id student1
uid=1001(student1) gid=1003(student1) groups=1003(student1),1001(interns)

(kali㉿kali)-[~]
$ id student2
uid=1002(student2) gid=1004(student2) groups=1004(student2),1001(interns)

(kali㉿kali)-[~]
$ id student3
uid=1003(student3) gid=1005(student3) groups=1005(student3),1002(cyberteam)

(kali㉿kali)-[~]
$
```

Part C: File Ownership

```
(kali㉿kali)-[~]
$ mkdir CyberSecurityProject

(kali㉿kali)-[~]
$ cd CyberSecurityProject

(kali㉿kali)-[~/CyberSecurityProject]
$ touch report.txt notes.txt credentials.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l
total 0
-rw-rw-r-- 1 kali kali 0 Jun 14 05:53 credentials.txt
-rw-rw-r-- 1 kali kali 0 Jun 14 05:53 notes.txt
-rw-rw-r-- 1 kali kali 0 Jun 14 05:53 report.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ sudo chown student1 report.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ sudo chown student1 report.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ sudo chown student1 notes.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l
total 0
-rw-rw-r-- 1 kali      kali 0 Jun 14 05:53 credentials.txt
-rw-rw-r-- 1 student1 kali 0 Jun 14 05:53 notes.txt
-rw-rw-r-- 1 student1 kali 0 Jun 14 05:53 report.txt

(kali㉿kali)-[~/CyberSecurityProject]
$
```

Original Owner : kali

New Owner : student1

Command Used: chown

Part D: File Permissions

```
Session  Actions  Edit  View  Help

(kali㉿kali)-[~/CyberSecurityProject]
$ touch security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l security_policy.txt
-rw-rw-r-- 1 kali kali 0 Jun 14 05:59 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ chmod 444 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l security_policy.txt
-r--r--r-- 1 kali kali 0 Jun 14 05:59 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ chmod 664 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l security_policy.txt
-rw-rw-r-- 1 kali kali 0 Jun 14 05:59 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ chmod 777 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$ ls -l security_policy.txt
-rwxrwxrwx 1 kali kali 0 Jun 14 05:59 security_policy.txt

(kali㉿kali)-[~/CyberSecurityProject]
$
```

chmod — Changes file permissions.

Permission	Symbolic Form	chmod Value
Read Only	r--r--r--	444
Read & Write	rw-rw-r--	664
Full Access	rwxrwxrwx	777

Part E: Permission Analysis

For the following permissions explain:

- 755
- 644
- 777
- 600
- 700

For each permission explain:

- Owner Rights
- Group Rights
- Other User Rights
- Real-world use case

Permission Analysis

1. Permission: 755 (rwxr-xr-x)

- Owner Rights: Read, Write, Execute
- Group Rights: Read, Execute
- Other User Rights: Read, Execute
- Real-World Use Case: Commonly used for directories and executable scripts where the owner can modify files, but others can only access and run them.

2. Permission: 644 (rw-r--r--)

- Owner Rights: Read, Write
- Group Rights: Read
- Other User Rights: Read
- Real-World Use Case: Used for normal text files and configuration files that the owner can edit while others can only view.

3. Permission: 777 (rwxrwxrwx)

- Owner Rights: Read, Write, Execute
- Group Rights: Read, Write, Execute
- Other User Rights: Read, Write, Execute
- Real-World Use Case: Gives full access to everyone. Generally not recommended because it creates security risks, but may be used temporarily for testing.

4. Permission: 600 (rw-----)

- Owner Rights: Read, Write
- Group Rights: No Access

- Other User Rights: No Access
- Real-World Use Case: Used for sensitive files such as passwords, private keys, and confidential documents.

5. Permission: 700 (rwx-----)

- Owner Rights: Read, Write, Execute
- Group Rights: No Access
- Other User Rights: No Access
- Real-World Use Case: Used for private directories and scripts that should only be accessible by the owner.

Part F: Security Challenge

Imagine you are a Linux Administrator.

Create a table showing:

File	Recommended Permission	Reason
password_backup.txt	600 (rw-----)	Contains sensitive password data and should only be accessible by the owner
public_notice.txt	644 (rw-r--r--)	Everyone can read the notice, but only the owner can modify it
system_log.txt	640 (rw-r-----)	The owner can edit the log, and authorized group members can read it
personal_notes.txt	600 (rw-----)	Personal information should remain private and accessible only to the owner

Part G: Linux Security Research

Research and explain:

1. Why are file permissions important?

File permissions help protect files from unauthorized access by deciding who can read, edit, or run a file.

2. What happens if sensitive files are given 777 permissions?

If a sensitive file has 777 permissions, everyone can read, change, or delete it. This is very risky because important data could be stolen or modified by anyone.

3. What is the principle of Least Privilege?

The Principle of Least Privilege means giving users only the access they need to do their work.

4. Why do organizations restrict user access?

Organizations restrict user access to protect important data and systems. It helps prevent mistakes, data leaks, and unauthorized actions by users.